MEAN VALUE THEOREM PRACTICE

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Name: Solutions

Theorem (Mean Value)**.** *Let f be a function that satisfies the following hypotheses:*

(1) f is continuous on the closed interval $[a, b]$.

(2) f is differentiable on the open interval (a, b).

Then there is a number c in (a, b) such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

or, equivalently,

$$f(b) - f(a) = f'(c)(b - a).$$

In Problems 1 through 4, verify that the function satisfies the hypotheses of the Mean Value Theorem on the given interval and find all numbers c that satisfy its conclusion.

1. $f(x) = x^3 - x^2 - 6x + 2$, $[0, 3]$

$f(3) = 27 - 9 - 18 + 2 = 27 - 27 + 2 = 2$

$f(0) = 2$

$\frac{f(3) - f(0)}{3 - 0} = \frac{2-2}{3} = 0$

$0 = f'(x) = 3x^2 - 2x - 6 \iff$

$x = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(3)(-6)}}{2(3)} = \frac{2 \pm \sqrt{4 + 4(18)}}{6}$

$= \frac{2 \pm \sqrt{4(19)}}{6} = \frac{2 \pm 2\sqrt{19}}{6} = \frac{1 \pm \sqrt{19}}{3}$

$4 < \sqrt{19} < 5 \Rightarrow \frac{5}{3} < \frac{1+\sqrt{19}}{3} < \frac{6}{3} = 2$ is in $(0,3)$

$-5 < -\sqrt{19} < -4 \Rightarrow \frac{-4}{3} < \frac{1-\sqrt{19}}{3} < \frac{-3}{3} = -1$ is not in $(0,3)$.

Therefore $\boxed{c = \frac{1+\sqrt{19}}{3}}$

2. $f(x) = \cos(2x)$, $[\pi/8, 7\pi/8]$

$f(7\pi/8) = \cos(7\pi/4) = \cos(7\pi/4 - 2\pi) = \cos(7\pi/4 - \frac{8\pi}{4}) = \cos(-\pi/4) = \cos(\pi/4) = f(\pi/8)$

$\Rightarrow \frac{f(7\pi/8) - f(\pi/8)}{7\pi/8 - \pi/8} = \frac{0}{6\pi/8} = 0.$

$0 = f'(x) = -2\sin(2x) \iff \sin(2x) = 0 \iff 2x = k\pi$, k any integer $\iff x = k\pi/2$

Since the only multiple of $\pi/2$ in $(\pi/8, 7\pi/8)$ is $\pi/2$, $\boxed{c = \pi/2}$

3. $f(x) = 3x^2 + 2x + 5$, $[-1, 1]$

$f(1) = 3 + 2 + 5 = 10$

$f(-1) = 3 - 2 + 5 = 6$

$\frac{f(1) - f(-1)}{1 - (-1)} = \frac{10 - 6}{2} = \frac{4}{2} = 2$

$2 = f'(x) = 6x + 2 \iff x = 0$

$\boxed{c = 0}$

4. $f(x) = \dfrac{x}{x+2}$, $[1, 4]$

$f(4) = \frac{4}{4+2} = \frac{4}{6} = 2/3$

$f(1) = \frac{1}{1+2} = \frac{1}{3}$

$\frac{f(4) - f(1)}{4 - 1} = \frac{1/3}{3} = 1/9$

$f'(x) = \frac{(x+2) - x}{(x+2)^2} = \frac{2}{(x+2)^2} = \frac{1}{9} \iff 18 = (x+2)^2$

$\iff \pm\sqrt{18} = x + 2$

$\iff x = -2 \pm \sqrt{18} =$

$4 < \sqrt{18} < 5 \Rightarrow 2 < -2 + \sqrt{18} < 3$

$-5 < -\sqrt{18} < -4 \Rightarrow -7 < -2 - \sqrt{18} < -6$

Therefore, the only solution in $(1,4)$ is $\boxed{c = -2 + \sqrt{18} = -2 + 3\sqrt{2}}$

5. Let $f(x) = 1 - x^{2/3}$. Show that $f(-1) = f(1)$, but there is no number c in $(-1, 1)$ such that

$$f'(c) = \frac{f(1) - f(-1)}{1 - (-1)} = \frac{0}{2} = 0.$$

Why does this not contradict the Mean Value Theorem?

$f'(x) = -2/3\, x^{-1/3} = \frac{-2}{3\sqrt[3]{x}} = 0$ has no solution, but this does not contradict the M.V.T. because $f'(0)$ is undefined; that is Hypothesis 2, which requires f to be differentiable on $(-1,1)$, is not satisfied.

6. Show that the equation $1 + 2x + x^3 + 4x^5 = 0$ has *exactly* one real solution.

Let $f(x) = x^5 + x^3 + 2x + 1$, so $f'(x) = 20x^4 + 3x^2 + 2 \geq 2$ for all x.

If there were two solutions, $s_1 < s_2$, then by the M.V.T. there would be a point $s_1 < c < s_2$ such that

$$f'(c) = \frac{f(s_2) - f(s_1)}{s_2 - s_1} = \frac{0 - 0}{s_2 - s_1} = \frac{0}{s_2 - s_1} = 0$$

which is clearly false. Hence there can only be at most one solution.

We can see one solution exists because f is continuous,

$$f(-1) = 1 - 2 - 1 - 4 = -6 < 0 \quad \text{and} \quad f(0) = 1 > 0$$

hence there exists some $-1 < s < 0$ such that $f(s) = 0$ by the Intermediate Value Theorem. Therefore $f(x) = 0$ has exactly one real solution.

7. Let c be a constant. Show that the equation $x^4 + 4x + c = 0$ has *at most* two real roots.

Let $f(x) = x^4 + 4x + c$, so $f'(x) = 4x^3 + 4 = 4(x^3 + 1) = 4(x+1)(x^2 - x + 1)$

Assume to the contrary that there exist three numbers $r_1 < r_2 < r_3$ such that $f(r_1) = f(r_2) = f(r_3) = 0$. By the Mean Value Theorem applied on (r_1, r_2) and (r_2, r_3), there exist numbers $r_1 < s_1 < r_2$ and $r_2 < s_2 < r_3$ such that

$$f'(s_1) = f'(s_2) = 0.$$

However, the discriminant of $x^2 - x + 1$ is $D = (-1)^2 - 4(1)(1) = 1 - 4 = -3 < 0$, hence $x^2 - x + 1$ has no real roots. That is, $x = -1$ is the only real solution to

$$0 = 4x^3 + 4 = 4(x+1)(x^2 - x + 1),$$

a contradiction. Therefore there are at most 2 solutions to $f(x) = 0$.